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In [19]: # Step 1: Basic Structure
def welcome_message():
    print("Welcome to the Library Management System")

# Initial library data structure with books, author, and availability status
library = {
    "Maths": {"author": "RS Agarwal", "available": True, "issued_to": None},
    "Physics": {"author": "RD sharma", "available": False, "issued_to": "Mitali"},
    "DSA": {"author": "Narasimha Karumuachi", "available": True, "issued_to": None},
    "Python": {"author": "Eric Matthes", "available": False, "issued_to": "Aniket"}
}

# Step 3: Display Books

def display_books():
    print("\nList of Books:")
    for title, details in library.items():
        status = "Available" if details["available"] else f"Issued to {details['issued_to']}"
        print(f"Title: {title}, Author: {details['author']}, Status: {status}")

# Step 2: Adding Books
def add_book():
    title = input("Enter the book title: ")
    author = input("Enter the author: ")
    if title in library:
        print("This book already exists in the library.")
    else:
        library[title] = {"author": author, "available": True, "issued_to": None}
        print(f"Book '{title}' added successfully.")

# Step 3: Issuing Books
def issue_book():
    title = input("Enter the book title to issue: ")
    if title in library and library[title]["available"]:
        person = input("Enter the name of the person issuing the book: ")
        library[title]["available"] = False
        library[title]["issued_to"] = person
        print(f"Book '{title}' issued to {person}.")
    else:

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        print("Book is either not available or does not exist.")

# Step 4: Returning Books
def return_book():
    title = input("Enter the book title to return: ")
    if title in library and not library[title]["available"]:
        library[title]["available"] = True
        library[title]["issued_to"] = None
        print(f"Book '{title}' returned successfully.")
    else:
        print("Book is either not issued or does not exist.")

# Step 5: Login System
credentials = {"admin": "admin123", "librarian": "lib123", "student": "student123*"}

def login():
    attempts = 3
    while attempts > 0:
        username = input("Enter username: ")
        password = input("Enter password: ")
        if username in credentials and credentials[username] == password:
            print("Login successful!!")
            return True
        else:
            attempts -= 1
            print(f"Invalid credentials. {attempts} attempts left.")
    return False

# Step 6: View Issued Books
def view_issued_books():
    print("\nIssued Books:")
    for title, details in library.items():
        if not details["available"]:
            print(f"Title: {title}, Issued to: {details['issued_to']}")

# Step 7: Main Menu
def main_menu():
    if not login():
        print("Access denied. Exiting system.")
    return
```

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while True:
    print("\nMain Menu:")
    print("1. View Books")
    print("2. Add a Book")
    print("3. Issue a Book")
    print("4. Return a Book")
    print("5. View Issued Books")
    print("6. Exit")

    choice = input("Enter your choice: ")
    if choice == '1':
        display_books()
    elif choice == '2':
        add_book()
    elif choice == '3':
        issue_book()
    elif choice == '4':
        return_book()
    elif choice == '5':
        view_issued_books()
    elif choice == '6':
        print("Exit the system. Goodbye!")
        break
    else:
        print("Invalid choice. Please try again.")

# Running the program
def run_library_management_system():
    welcome_message() # Display the welcome message
    main_menu() # Start the menu-driven system

# Entry point for the script
if __name__ == "__main__":
    run_library_management_system()
```

Welcome to the Library Management System

Login successful!!

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

List of Books:

Title: Maths, Author: RS Agarwal, Status: Available
Title: Physics, Author: RD sharma, Status: Issued to Mitali
Title: DSA, Author: Narasimha Karumuachi, Status: Available
Title: Python, Author: Eric Matthes, Status: Issued to Aniket

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Book 'DSA' issued to AKANKSHA.

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Book 'CHEMISTRY' added successfully.

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Issued Books:

Title: Physics, Issued to: Mitali

Title: DSA, Issued to: AKANKSHA

Title: Python, Issued to: Aniket

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Book 'DSA' returned successfully.

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

List of Books:

Title: Maths, Author: RS Agarwal, Status: Available

Title: Physics, Author: RD sharma, Status: Issued to Mitali

Title: DSA, Author: Narasimha Karumuachi, Status: Available

Title: Python, Author: Eric Matthes, Status: Issued to Aniket

Title: CHEMISTRY, Author: S.CHAND, Status: Available

Main Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Exit the system. Goodbye!

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